

Robotics

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This is a set of notes covering interesting concepts I find while studying the Modern Robotics textbook. I will try to bring analytic theory whenever applicable.

Textbooks Used to Develop these Notes:

1. *Modern Robotics* - Kevin Lynch, Frank Park
2. *Calculus on Manifolds* - Michael Spivak

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NOTE: Any sections/subsections marked with an asterisk are extra sections and have been added to this document as extensions of introduced theory.

1 Holonomic Constraints

1.1 Basic Multivariable Analysis

Note: The following notes may not be entirely rigorous! I have attempted to write them logically to the best of my abilities so that they follow a structured narrative.

We begin with the exploration of multivariate analysis. When we deal with a multivariate function in $\mathbb{R}$, we deal with some map of the form $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We can define this more formally via vector operations as the following:

$$f \left(\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n) \\ \vdots \\ f_m(x_1, x_2, \dots, x_n) \end{bmatrix} \quad (1)$$

such that $\forall j \in [1, m] : f_j : \mathbb{R}^n \rightarrow \mathbb{R}$. Our original definition of the derivative in $\mathbb{R}$ was:

$$f'(x) := \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

or equivalently

$$\lim_{z \rightarrow x} \frac{f(z) - f(x) - f'(x)(z - x)}{z - x} = 0$$

Using the substitution $h = z - x$, we also get the equivalent definition:

$$\lim_{h \rightarrow 0} \frac{f(x + h) - f(x) - f'(x)h}{h} = 0$$

In which case, our $\epsilon - \delta$ definition would be:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{|f(x + h) - f(x) - f'(x)h|}{|h|} < \epsilon$$

However if we try to plainly substitute the definition for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ into the first definition, we get "division of two vectors" which is not a defined operation for vector spaces (they are only abelian groups under addition).

We consider the following lemma in order to understand how to extend the derivative to the multivariate case:

Lemma 1.1. $f'(x)$ exists $\iff \left\{ \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| \right\}_{\delta > 0} \rightarrow 0$

Proof. $\implies$: If $f'(x)$ exists, then:

$$\begin{aligned} \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \implies |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot |z - x| < \epsilon \cdot \delta \end{aligned}$$

Because this is true for all z in the punctured δ -ball around x , we have an upper bound, so we take the supremum:

$$\implies \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

And so we rearrange the terms to get a δ -net that converges such that:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \frac{1}{\delta} \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon$$

Thus proving the forward direction.

$\Leftarrow$: Now for the converse, consider:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0 : \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta$$

Now for any arbitrary δ -ball around x where $\delta \leq \delta_0$, we have that:

$$\begin{aligned} & \sup_{z \in B(x, \delta) \setminus \{x\}} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \cdot \delta \\ \implies & \forall z \in B(x, \delta) \setminus \{x\} : \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \\ \implies & \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \frac{|f(z) - f(x) - f'(x)(z - x)|}{|z - x|} < \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon \end{aligned}$$

Since δ was arbitrary as long as it was less than or equal to δ_0 , we observe:

$$\begin{aligned} & \forall \epsilon > 0, \exists \delta_0 > 0, \forall \delta \leq \delta_0, \forall z \in B(x, \delta_0) \setminus B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in \bigcup_{\delta \in (0, \delta_0]} (B(x, \delta_0) \setminus B(x, \delta)) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \\ \iff & \forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \bigcap_{\delta \in (0, \delta_0]} B(x, \delta) : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon \end{aligned}$$

Now because $\mathbb{R}$ is a T_1 space, we see that the intersection in the previous statement is equivalent to $\{x\}$, so we get:

$$\forall \epsilon > 0, \exists \delta_0 > 0, \forall z \in B(x, \delta_0) \setminus \{x\} : \left| \frac{f(z) - f(x)}{z - x} - f'(x) \right| < \epsilon$$

A useful equivalent statement is:

$$\forall \epsilon > 0, \exists \delta_0 < \epsilon, \forall \delta \leq \delta_0, \forall z \in B(x, \delta) \setminus \{x\} : \frac{1}{\delta} |f(z) - f(x) - f'(x)(z - x)| < \epsilon$$

□

Now essentially this supremum δ -net, similar to oscillation is the worst case net. The important part of this result is the idea of linear approximation, such that:

$$f'(x)(z - x) + f(x) - \epsilon^2 < f(z) < f'(x)(z - x) + f(x) + \epsilon^2$$

or as we write it in calculus $f(z) \approx f(x) + f'(x)(z - x)$, which is the tangent line approximation.

We can apply this approximation idea to the multivariate case, but first we introduce the following notion:

Definition 1.1. Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $i \in [1, m], j \in [1, n]$:

$$\frac{\partial f_i(x)}{\partial e_j} := \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x)}{h}$$

where $\{e_1, \dots, e_n\}$ is the canonical basis of $\mathbb{R}^n$. This is the definition of the partial derivative.

Since each $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, we see that the partial derivative is scalar valued. We now show that it admits the linear approximation property. We can rewrite our limit definition as:

$$\lim_{h \rightarrow 0} \left(\frac{f_i(x + e_j h) - f_i(x)}{h} - \frac{\partial f_i(x)}{\partial e_j} \right) = \lim_{h \rightarrow 0} \frac{f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h}{h} = 0$$

So with an argument similar to Lemma 1.1, we see that $f_i(z) \approx f_i(x) + \left(\frac{\partial f_i}{\partial e_j} \right) h$. Now observe that:

$$f(z) := \begin{bmatrix} f_1(z) \\ \vdots \\ f_m(z) \end{bmatrix}$$

Lemma 1.2. For all $i \in [1, m]$ and some $x \in \mathbb{R}^n$ and an e_j :

$$\frac{\partial f_i(x)}{\partial e_j} \text{ exist}$$

if and only if:

$$\frac{\partial f(x)}{\partial e_j} \text{ exists}$$

Proof. $\implies$: We claim that the partial derivative of f takes the following form:

$$\frac{\partial f(x)}{\partial e_j} = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_j} \\ \vdots \\ \frac{\partial f_m(x)}{\partial e_j} \end{bmatrix}$$

We have some δ_i given by the existence of each partial derivative, so we define $\delta_0 := \min\{\delta_i : 1 \leq i \leq n\}$. Then for all $h \in B(0, \delta_0)$:

$$\frac{\|f(x + e_j h) - f(x) - \left(\frac{\partial f(x)}{\partial e_j} \right) h\|}{|h|} \leq \frac{1}{|h|} \sum_{i=1}^m \left| f_i(x + e_j h) - f_i(x) - \left(\frac{\partial f_i(x)}{\partial e_j} \right) h \right| < m\epsilon$$

where we used the inequality that the $\|v\| \leq \sum |v_i|$.

$\Leftarrow$: We claim that:

$$\frac{\partial f_i(x)}{\partial e_j} = e_i \cdot \frac{\partial f(x)}{\partial e_j}$$

We now use the inequality $|v_i| \leq \|v\|$ and find that $\forall i \in [1, m]$:

$$\frac{|f_i(x + e_j h) - f_i(x) - (e_i \cdot \frac{\partial f(x)}{\partial e_j}) h|}{|h|} \leq \frac{\|f(x + e_j h) - f(x) - \left(\frac{\partial f(x)}{\partial e_j} \right) h\|}{|h|} < \epsilon$$

so the same δ suffices.

Now once again with an argument similar to Lemma 1.1, we can extend the idea of linear approximation and say that for all z such that $z = x + e_j h$:

$$f(z) \approx f(x) + \left(\frac{\partial f(x)}{\partial e_j} \right) h$$

□

Notice that the proofs did not use any properties of the basis vector e_j itself, so we can extend the definition:

Definition 1.2. Given some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ in the form of equation (1), then for some $v \in \mathbb{R}^n$ and $i \in [1, m], j \in [1, n]$:

$$\frac{\partial f_i(x)}{\partial v} := \lim_{h \rightarrow 0} \frac{f_i(x + hv) - f_i(x)}{h}$$

and similarly for f . Such a derivative is called the Gateaux derivative.

We can extend the Gateaux derivative to obtain the Frechet derivative. The Gateaux derivative limited itself to a specific direction of the vector v (and then used h as the variable for the limit).

Definition 1.3. The Frechet derivative of a function f at $x \in \mathbb{R}^n$ is:

$$Df(x) : \lim_{v \rightarrow 0} \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} = 0$$

or equivalently:

$$\forall \epsilon > 0, \exists \delta > 0, \forall v \in B(0, \delta) : \frac{\|f(x + v) - f(x) - (Df(x))(v)\|}{\|v\|} < \epsilon$$

where the norm induces a metric so that $v \in B(0, \delta)$ (in this case 0 is the 0 vector), is the same as saying $\|v\| \in B(0, \delta)$ or equivalently $\|v\| < \delta$.

The form of the limit is similar to the idea of linear approximation. Since $v \in \mathbb{R}^n$, then we must have that $Df(x) \in \mathcal{M}_{m \times n}(\mathbb{R})$. If the function is Frechet differentiable, then $Df(x)$ is the Jacobian:

$$Df(x) = \begin{bmatrix} \frac{\partial f_1(x)}{\partial e_1} & \dots & \frac{\partial f_1(x)}{\partial e_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x)}{\partial e_1} & \dots & \frac{\partial f_m(x)}{\partial e_n} \end{bmatrix} = \begin{bmatrix} \frac{\partial f(x)}{\partial e_1} & \dots & \frac{\partial f(x)}{\partial e_n} \end{bmatrix}$$

Proof. If we chose the family of vectors $\{he_j : h \in \mathbb{R} \wedge j \in [1, n]\}$, then differentiability tells us:

$$\forall \epsilon > 0, \exists \delta > 0, \forall h \in B(0, \delta) : \frac{\|f(x + he_j) - f(x) - h(Df(x))e_j\|}{|h|} < \epsilon$$

telling us that:

$$\frac{\partial f(x)}{\partial e_j} = Df(x)e_j$$

which is the j -th column of the Jacobian matrix. □

Notice that Lemma 1.2 was a statement on equivalence; however, the converse is not necessarily true for the Jacobian matrix. The total differentiability of f at x gives us the existence of partial derivatives, but not

necessarily the converse.

Consider the function:

$$f(x, y) = \frac{\sqrt{xy}}{(x + y)^2}$$

when $x \neq 0$ and 0 when $x = 0$.

1.2 Inverse and Implicit Function Theorems

1.3 Configuration Spaces